INFOSYS MOCK TEST 3

Hard

Rules and Instructions

- Attempt every question within the suggested time.
- Do not use calculators, notes, search engines or AI.
- Select one best answer for every MCQ.
- No negative marking is assumed for this practice test.
- Coding questions require efficient solutions and complexity analysis.
- Explain your reasoning during technical and HR interviews.
- These are original practice questions, not an official Infosys examination paper.

ROUND 1: ONLINE ASSESSMENT

Section A: Logical Reasoning

10 Questions | 15 Minutes

1. Find the next term: 2, 6, 15, 31, 56, ...
   A. 82 B. 88
   C. 92 D. 96

2. If the letters of COMPUTER are arranged alphabetically, which letter is fourth?
   A. E B. M
   C. O D. P

3. In a code, each letter is replaced by the letter three places ahead, wrapping after Z. Encode LOGIC.
   A. ORJLF B. ORJID
   C. MPHJD D. ORJLC

4. A, B, C, D and E sit in a row. A is not at an end. B is immediately left of C. D is at the right end. E is not next to D. Which arrangement is possible?
   A. A B C E D B. B C A E D
   C. E A B C D D. C B A E D

5. A is older than B but younger than C. D is younger than B. E is older than C. Who is second oldest?
   A. A B. B
   C. C D. D

6. A survey of 100 people finds 60 use app A, 50 use app B and 30 use both. How many use exactly one app?
   A. 40 B. 50
   C. 60 D. 80

7. If 5 machines make 5 parts in 5 minutes, how long will 100 machines take to make 100 parts?
   A. 5 minutes B. 20 minutes
   C. 100 minutes D. 500 minutes

8. A cube is painted on all faces and cut into 27 equal smaller cubes. How many small cubes have exactly two painted faces?
   A. 8 B. 12
   C. 16 D. 20

9. A person travels half a journey at 40 km/h and the other half at 60 km/h. Average speed is:
   A. 45 km/h B. 48 km/h
   C. 50 km/h D. 52 km/h

10. Some APIs are secure. All secure systems are monitored. Which conclusion follows?
   A. All APIs are monitored B. Some APIs are monitored
   C. No API is monitored D. All monitored systems are APIs

Section B: Mathematical Ability

10 Questions | 25 Minutes

1. If $x - \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$.
   A. 7 B. 9
   C. 11 D. 13

2. A sum doubles in 8 years at simple interest. In how many years will it become five times?
   A. 24 B. 28
   C. 32 D. 36

3. A and B can complete a task in 12 and 18 days. They work together for 4 days, then A leaves. How many additional days does B need?
   A. 6 B. 8
   C. 10 D. 12

4. How many distinct arrangements can be made from the letters of LEVEL?
   A. 20 B. 30
   C. 60 D. 120

5. A box contains 4 defective and 6 good components. Two are selected without replacement. Probability both are good is:
   A. 1/3 B. 2/5
   C. 1/2 D. 3/5

6. If the roots of $t^2 - 7t + 10 = 0$ are a and b, find $a^2 + b^2$.
   A. 19 B. 29
   C. 39 D. 49

7. A mixture of milk and water is 5:3. If 16 litres of mixture is removed and replaced with water from a 64-litre vessel, what is the new milk:water ratio?
   A. 5:11 B. 3:5
   C. 1:2 D. 5:7

8. A person invests Rs. 20,000 at 8% compound interest annually. Amount after 2 years is:
   A. Rs. 22,800 B. Rs. 23,000
   C. Rs. 23,328 D. Rs. 24,000

9. A circle has radius 7 cm. Using $\pi = 22/7$, its area is:
   A. 144 cm^2 B. 154 cm^2
   C. 164 cm^2 D. 176 cm^2

10. A data set has mean 20 for 5 values. If one value 12 is replaced by 22, the new mean is:
   A. 20 B. 21
   C. 22 D. 23

Section C: Verbal Ability

10 Questions | 12 Minutes

1. Choose the synonym of Pragmatic.
   A. Idealistic B. Practical
   C. Uncertain D. Emotional

2. Choose the antonym of Concur.
   A. Agree B. Approve
   C. Disagree D. Confirm

3. Choose the grammatically correct sentence.
   A. The data is analysed by the team yesterday. B. The data were analysed by the team yesterday.
   C. The data were analyse by the team yesterday. D. The data has analyse yesterday.

4. No sooner had the meeting started _____ the power failed.
   A. when B. then
   C. than D. that

5. The proposal was rejected, not because it was costly, _____ because it lacked evidence.
   A. and B. but
   C. so D. or

6. Choose the correct reported speech: He said, "I am working."
   A. He said that he is working. B. He said that he was working.
   C. He says he was working. D. He told that I am working.

7. Choose the word closest to Resilient.
   A. Fragile B. Adaptable
   C. Rigid D. Indifferent

8. Identify the error: The committee have submitted its report.
   A. committee B. have
   C. its D. report

9. Choose the correctly spelt word.
   A. Entrepreneur B. Enterpreneur
   C. Entreprenuer D. Entreprenur

10. The code is concise; _____, it is difficult to maintain.
    A. moreover B. however
    C. therefore D. similarly

Section D: Pseudocode and Output Tracing

5 Questions | 10 Minutes

1. Output?
   `f(n):  if n==0 return 0; return n+f(n-1); print(f(4))`
   A. 6 B. 10
   C. 16 D. 24

2. Output?
   `a=[3,1,4,1]; for i=1 to 3:  if a[i]<a[i-1]:  swap(a[i],a[i-1]); print(a[0])`
   A. 1 B. 3
   C. 4 D. 1,3

3. A binary search is performed on a sorted array of 1024 elements. Maximum comparisons in the usual worst-case order is approximately:
   A. 10 B. 11
   C. 512 D. 1024

4. What is the time complexity?
   `for i=1 to n:  j=1; while j<n:  j=j*2`
   A. $O(n^2)$ B. $O(n \log n)$
   C. $O(\log n)$ D. $O(2^n)$

5. Output?
   `x=1; for i=1 to 4:  x=2*x+1; print(x)`
   A. 15 B. 16
   C. 31 D. 32

Section E: Puzzle Solving

4 Questions | 8 Minutes

1. You have 12 coins, one counterfeit coin that may be heavier or lighter. Describe a strategy to identify the counterfeit coin and determine whether it is heavier or lighter using at most 3 balance-scale weighings.

2. Three people make statements. A says B is lying. B says C is lying. C says A and B are both lying. Each person is either always truthful or always lying. Determine who is truthful.

3. A sequence of tasks has durations 2, 3, 5, 7 and 11 units. Two processors can run one task each at a time. Give a schedule that minimises completion time and state the makespan.

4. There are 25 horses and 5 lanes. You can race only 5 horses at a time and have no stopwatch. What is the minimum number of races needed to identify the three fastest horses? Explain the elimination logic.

Section F: English Writing

10 Minutes
Write 200–250 words on:
Artificial intelligence will change the role of entry-level software engineers. Discuss opportunities, risks and the skills graduates should develop.

ROUND 2: ADVANCED CODING ASSESSMENT

Suggested Time: 90 Minutes

1. **Minimum Number of Coins**

   Given coin denominations and a target amount, return the minimum number of coins required to form the amount, or -1 if impossible.

   Each denomination may be used unlimited times.

   Example:

   Coins: `[1,2,5]`

   Amount: 11

   Output: `3`

   Target: Dynamic programming, O(nA).

2. **Number of Islands**

   Given an m x n grid of 0s and 1s, count connected groups of 1s using four-directional adjacency.

   Target: DFS/BFS, O(mn).

3. **Longest Increasing Subsequence**

   Given an integer array, return the length of its longest strictly increasing subsequence.

   Target: O(n log n) using binary search.

   Also explain the O(n squared) dynamic programming approach.

ROUND 3: TECHNICAL INTERVIEW

1. Explain abstraction, encapsulation, inheritance and polymorphism with examples from your own code.
2. What are SOLID principles? Explain any two with practical examples.
3. Explain Java memory areas: stack, heap and method area/metaspace.
4. What is the Java Collections Framework?
5. Compare HashMap, TreeMap and LinkedHashMap.
6. Explain the contract between equals() and hashCode().
7. What is a race condition? How can synchronisation help?
8. Explain deadlock, starvation and livelock. How are they different?
9. Explain BFS and DFS implementation and complexity.
10. Explain dynamic programming. How do memoization and tabulation differ?
11. What is the difference between greedy algorithms and dynamic programming?
12. Give an example where greedy fails.
13. Explain binary search on answer with a suitable example.
14. What is a trie? Where is it useful?
15. Explain normalization up to 3NF.
16. What problems can denormalization solve?
17. Compare B-tree and B+ tree indexes at a high level.
18. Write SQL to find duplicate email addresses in a Users table.
19. Write SQL to return the department with the highest average salary.
20. Explain transaction isolation levels and a dirty read.
21. What is the CAP theorem?
22. Explain load balancing, caching and horizontal scaling.
23. Design a basic URL-shortening service. Discuss API endpoints, database fields, unique-code generation and scaling.
24. What happens when a user enters a URL in a browser?
25. Explain authentication versus authorization.
26. What are embeddings and retrieval-augmented generation (RAG)?

27. Explain training, fine-tuning and inference in machine learning.

28. Deep-dive your strongest project: architecture, data flow, schema, security, performance, trade-offs, failures, testing, deployment and your personal contribution.

ROUND 4: HR, BEHAVIOURAL AND MANAGERIAL INTERVIEW

1. Give a concise introduction connecting your education, skills and strongest project to the role.

2. Why should Infosys select you over another fresher with similar marks?

3. Describe a time you influenced a team without formal authority.

4. Tell me about a disagreement over a technical decision. How did you reach a resolution?

5. Describe a situation where you had to learn a new tool quickly.

6. How would you explain a technical delay to a non-technical client?

7. Suppose a client requests a feature you believe is unsafe or unreliable. What would you do?

8. What would you do if assigned to a project using a language you have never used?

9. How do you respond when your manager gives you critical feedback?

10. Describe a time you handled multiple deadlines. What did you prioritise and why?

11. Are you comfortable with relocation, training, shifts or project allocation as stated in the official offer?

12. What are your short-term learning goals?

13. What does professional integrity mean to you?

14. If you discover a teammate committed a serious defect just before release, what would you do?

15. Do you have questions about the role, training, team or expectations?

ANSWER KEY

Logical Reasoning:
1-C, 2-B, 3-A, 4-A, 5-A, 6-D, 7-A, 8-B, 9-B, 10-B.
Mathematical Ability:
1-C, 2-C, 3-C, 4-B, 5-C, 6-B, 7-A, 8-C, 9-B, 10-B.
Verbal Ability:
1-B, 2-C, 3-B, 4-C, 5-B, 6-B, 7-B, 8-B, 9-A, 10-B.
Pseudocode:
1-B, 2-A, 3-B, 4-B, 5-C.
Puzzle Guidance:

1. This is the classic 12-coin, three-weighing problem. A full-credit answer must construct a decision tree that uniquely distinguishes all 24 possibilities (12 coins, each possibly heavier or lighter).

2. A and C are truthful; B is lying.

3. Total work is 28 units. Schedule 11+3 on one processor and 7+5+2 on the other. Both processors finish in 14 units, so the makespan is 14.

4. Standard solution: 7 races. Race five groups of five horses, then race the five group winners. Use the resulting order to eliminate horses that cannot be in the top three, then race the remaining plausible candidates.